Zero-Knowledge Federated Learning with Lattice-Based Hybrid Encryption for Quantum-Resilient Medical AI

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***Abstract*—Hospital federated learning (FL) must simultaneously resist gradient inversion, Byzantine poisoning, and Harvest-Now-Decrypt-Later (HNDL) risk under multi-year retention. ZKFL-PQ layers four mechanisms with explicit binding: NIST ML-KEM-768 transport; an Unruh NIZK (library default r=128) of ℓ_2-bounded updates in language $\mathcal{L}_\tau^{\text{bind}}$, plus an Enc-consistency gadget for BFV coins; full-vector BFV aggregation opened by (t, n)-threshold *partial* decryption (the server secret sk is never reconstructed); and a default post-ZKP coordinate-wise median against sign-flip and in-bound backdoors that pure ℓ_2 gates must accept. A public artifact ships fused SEAL+threshold HE, shared-SIS Unruh, bit-level Keccak/SHA3 formal checks, Lean/Python combinatorial Unruh bounds, and measured payloads under varying r and HE degree. Empirically we report multi-seed synthetic baselines, scale N=20/T=30, UCI Breast Cancer, full-resolution PneumoniaMNIST (MLP and `ConvNet28`), and backdoor attack success rates for ZKP+median vs. FedAvg/ℓ_2-only. ZKFL-PQ is complementary to Beskar-class secure aggregation rather than a drop-in replacement.**

***Index Terms*—Post-quantum cryptography, federated learning, zero-knowledge proofs, homomorphic encryption, medical AI, Byzantine robustness**

I. Introduction

Federated learning (FL) [1] keeps raw records at the edge, yet model updates remain vulnerable to gradient inversion [2], Byzantine poisoning [3], and HNDL attacks against classical TLS when medical archives outlive RSA/ECC [4]. Hospital consortia therefore need (i) post-quantum (PQ) session confidentiality, (ii) verifiable rejection of oversized updates, and (iii) aggregation that does not place a single curious server in sole possession of plaintext gradients.

Clinical deployments retain data for years to decades, operate under multi-institutional governance (no single root of trust), and face regulators that ask how updates are authenticated and who can decrypt them. Wrapping FedAvg in classical TLS is insufficient against adversaries that store ciphertexts today and open them later [4]. Cryptographic accept alone is also insufficient: updates whose ℓ_2 norm respects a public threshold τ can still flip signs or implant triggers, so statistical robust aggregation must follow a successful proof.

a) Design goals.: We optimize for hospital consortia where (G1) ciphertext retention is measured in years, (G2) no single IT operator may hold a usable sk, (G3) oversized gradients must be rejectable without trusting the client, and (G4) in-bound Byzantine behavior that passes τ is handled by a robust aggregator rather than by tightening τ until honest sites fail. These goals motivate the four-layer stack rather than any single primitive.

b) Contributions.:

1) A layered protocol that binds Unruh norm proofs to BFV ciphertexts via $\mathcal{L}_\tau^{\text{bind}}$ + Enc-consistency, opens with threshold partial decrypt, then robust-aggregates (default median), with each layer addressing a distinct threat (Table I).
2) Concrete Unruh (r=128) and BFV parameters, a named soundness/privacy theorem, Lean/Python/`QROM.ec` combinatorial checks with SHA3/Keccak lane lemmas, plus measured payloads under varying r and HE degree.
3) A reproducible artifact [12]: full-vector encrypt+prove on a shared dimension, crypto-only accept, fused SEAL+threshold path, shared SIS commitment across Unruh sessions, no monolithic server sk.
4) An empirical study spanning large-norm and sign-flip attacks, backdoor ASR with ZKP+median (Fig. 3), UCI Breast Cancer, full-resolution MedMNIST (MLP and `ConvNet28`), and scale N=20, T=30.

II. Threat Model and Claims

Adversary: (i) a network eavesdropper with future quantum computers against factoring/DLP; (ii) up to $f<N/2$ Byzantine clients that may submit arbitrary vectors and proofs; (iii) a curious aggregator that sees all protocol messages but does not hold sk alone; (iv) up to $t-1$ colluding threshold decryptors.

Trust. Honest clients follow local SGD and the Prove interface. Decryptors follow the partial-decrypt algorithm. The aggregator correctly runs Verify and the chosen robust aggregator; it need not be trusted for confidentiality of individual updates once shares are distributed.

Claims: ML-KEM-768 transport (FIPS 203 Level 3) [5]; norm soundness for oversized witnesses under SIS + (Unruh/)RO assumptions; ciphertext binding so a proof cannot

TABLE I
THREAT COVERAGE BY LAYER (COMPOSITION IS THE CONTRIBUTION).

Threat	ML-KEM	ZKP ℓ_2	Enc-cons.	Thr. open	Median
HNDL / PQ transit	yes	–	–	–	–
Oversized update	–	yes	–	–	–
Proof/ct split	–	bind	yes	–	–
Curious aggregator	–	–	–	yes	–
Sign-flip / backdoor	–	no‡	–	–	yes

‡In-bound updates satisfy τ; Table IV reports 0% crypto detection.

TABLE II
POSITIONING VS. RELATED PQ / ROBUST FL LINES.

System	PQ transport	Norm ZKP	HE / thr. dec.
Beskar [8]	yes (masking/DP)	–	secure agg. stack
Patel [9]	lattice HE	–	HE FL
Lu et al. [10]	hybrid PQ	–	HE FL
Krum [3]	statistical	–	–
zkML [11]	–	inference ZK	–
ZKFL-PQ (ours)	ML-KEM-768	Unruh+bind	BFV(t, n)

be replayed on a different $\mathbf{ct}$; aggregator privacy via Shamir-shared sk and partial decryption $\mu_i = c_1 \star (\lambda_i \cdot s_i)$.

Scope. We do not claim differential privacy or a drop-in replacement for Beskar-class full-stack secure aggregation [8]. ZKFL-PQ targets verifiable norm gating, multi-party decrypt, and robust post-filtering; pure ℓ_2 alone does not stop sign-flip or in-bound backdoors (Table I).

III. RELATED WORK

Beskar offers a mature PQ secure-aggregation stack with assisting nodes and DP [8]. Patel [9] and Lu et al. [10] study PQ-HE FL without ciphertext-bound norm ZKPs. zkML targets inference proofs rather than FL aggregation [11]. Multi-Krum remains effective against low-norm adversaries; ZKFL-PQ therefore composes cryptographic accept with median/Krum rather than relying on ℓ_2 alone. We report Hybrid+mean vs. Hybrid+median to make this separation measurable (Table IV).

IV. PROTOCOL DESIGN

A. Round Structure

Each round t: the server broadcasts $w^{(t)}$. Client i runs local SGD on D_i, quantizes to $\tilde{w}_i$, encrypts under BFV public key pk, proves $\|\tilde{w}_i\|_2 \le \tau$ bound to $\mathbf{ct}_i$ (and Enc-consistency of coins), and sends an ML-KEM-wrapped payload. The server verifies proofs, threshold-opens each accepted ciphertext, and applies coordinate-wise median (default) before the model step (Algorithm 1).

a) Transport.: ML-KEM-768 encapsulates a session key used to AES-CTR-wrap $(\mathbf{ct}_i, \pi_i)$. This protects in-transit confidentiality against classical and future quantum adversaries on the channel; it does not replace HE confidentiality against the aggregator, which follows from RLWE + threshold opening.

b) Verification checklist.: $\mathsf{Verify}(\pi_i, \mathbf{ct}_i, \tau)$ returns 1 iff: (i) Unruh invertible-RO records are consistent; (ii) Fiat–Shamir/Unruh challenge bits match the recomputed digest of first messages and $\mathbf{ct}_i$; (iii) each session satisfies the SIS algebraic relation against the shared $\mathbf{A}$; (iv) response norms respect the rejection bound B; (v) Enc-consistency accepts for every BFV chunk. Fail-closed: any failed check drops the client for that round.

B. Quantization

Quantize clips coordinates to $[-c, c]$, scales by s, and rounds into $\mathbb{Z}_t$ (artifact defaults c=1, s=10^3). The public threshold τ applies after quantization; adaptive rules $\tau^{(t)} = \hat{\mu} + k\hat{\sigma}$ are compatible. Reported runs use fixed τ (synthetic 5; medical 8–25 depending on model scale). Quantization is part of the proven statement: the ZKP and HE operate on the same $\tilde{w}$, so a client cannot prove a compact vector while encrypting a different full-precision update.

C. Bound Norm Language

Definition 1 (Bound language). *Let* $C = \mathsf{Commit}(\tilde{w}; \mathbf{r}) = \mathbf{A}(\tilde{w}\|\mathbf{r}) \bmod q'$ *and* $\mathbf{ct} = \mathsf{Enc}_{pk}(\tilde{w}; \rho)$. *Then*

$$\begin{aligned}\mathcal{L}_\tau^{\text{bind}} = \big\{ &(C, \mathbf{ct}, \tau) : \exists (\tilde{w}, \mathbf{r}, \rho) \\ &C = \mathsf{Commit}(\tilde{w}; \mathbf{r}),\ \mathbf{ct} = \mathsf{Enc}(\tilde{w}; \rho),\ \|\tilde{w}\|_2 \le \tau \big\}. \quad (1)\end{aligned}$$

a) Σ-protocol [6].: Commit masks $\mathbf{y}$ and message $\tilde{w}$; the Fiat–Shamir challenge is $e = H(C\|\mathbf{ct}\|T\|\tau)$; responses $(\mathbf{z}, \mathbf{r}_z)$ satisfy $T + eC \equiv \mathbf{A}(\mathbf{z}\|\mathbf{r}_z)$ and $\|\mathbf{z}\|_2 \le B$. Rejection sampling yields HVZK in the classical ROM. Acceptance uses only algebraic and norm checks—never a client-supplied Boolean.

b) Unruh-oriented NIZK.: Classical FS is not tightly QROM-secure [7]. We use parallel binary Unruh sessions (library default r=128; latency-sensitive demos often 4–64) with invertible-RO records $(\rho, H(\rho))$. Challenge bits are extracted from $H(\mathsf{UNRUH}\|\tau\|\mathbf{ct}\|\mathsf{first\text{-}msgs})$. All sessions share one SIS matrix $\mathbf{A}$ (not r independent keys), which keeps prove/verify interactive at imaging-scale d (one $256\times(d+128)$ matvec family instead of r separate keys). Combinatorial uniqueness / 2^{-r} lemmas are machine-checked in Lean 4; executable game hops and a SHA3-instantiated EasyCrypt QROM library with bit-level Keccak-f[1600] lane lemmas (θ-column parity, ρ/π/χ/ι locality, packing invertibility, 24-round fold) accompany the artifact (`formal/run_formal_ci.py`).

c) Enc-consistency.: Beyond hashing $\mathbf{ct}$ into the challenge, a dedicated Σ-gadget proves knowledge of coins $\rho = (u, e_0, e_1)$ such that the BFV encryption equations hold for the same plaintext chunk, closing split attacks that pair a benign proof with a mismatched ciphertext. In the artifact, Enc-consistency is proved per chunk and bound into the same associated-data digest as the Unruh sessions.

TABLE III
CRYPTOGRAPHIC PARAMETERS (RESEARCH STACK).

Component	Setting	Notes
ML-KEM-768	n=256, k=3, q=3329	FIPS 203 L3 [5]
BFV degree	n=512 / 4096	Lightweight vs. `classic128`
BFV moduli	$q{\approx}2^{40}$, $t{=}2^{12}$	NumPy encoder; optional estimator / SEAL
Threshold	$(t,n){=}(2,3)$ partial decrypt	Server $sk{=}\perp$
Unruh reps	r=128 default	Latency-sensitive runs may use r=4–64
Post-ZKP agg	median (default)	Multi-Krum / mean overrides
Packing	full-vector chunks	All d coordinates encrypted
SIS commit	shared $\mathbf{A}$	One $256{\times}(d{+}128)$ matrix / prover

TABLE IV
MEAN±STD OVER 5 SEEDS (SYNTHETIC). ACCURACY (%) AND DETECTION.

Method	Acc. (large)	Acc. (flip)	Det. (large)	Det. (flip)
FedAvg	26.0±1.0	41.0±13.2	0%	0%
Clip@τ	100±0	40.3±12.7	–	–
Multi-Krum	100±0	100±0	–	–
HE-only	22.1±2.1	19.8±9.5	0%	0%
ZKP-only	84.9±30.2	40.9±13.2	97%	0%‡
Hybrid+mean	87.0±26.0	40.1±12.6	97%	0%‡
Hybrid+median	100±0	100±0	97%	0%‡

‡Sign-flip respects τ—ℓ_2 proofs must accept (0% crypto detection). Hybrid+mean keeps plaintext mean after accept; Hybrid+median is the default composition and restores accuracy [12].

Algorithm 1 ZKFL-PQ round (default composition)

Require: $w^{(t)}$, τ, KEM (ek_S, dk_S), HE pk, decryptor shares
Broadcast $w^{(t)}$
for each client i **do**
  $\tilde{w} \leftarrow \mathsf{Quantize}(\mathsf{LocalSGD}(w^{(t)}, D_i))$
  $\mathbf{ct}_i \leftarrow \mathsf{Enc}_{pk}(\tilde{w};\rho_i)$; prove Enc-consistency of ρ_i
  $\pi_i \leftarrow \mathsf{Prove}_{\mathsf{Unruh}}((\mathbf{ct}_i,\tau),\tilde{w})$
  send ML-KEM-wrapped $(\mathbf{ct}_i, \pi_i)$
end for
Keep i iff cryptographic verify returns 1
For each accepted i: threshold-open $\mathbf{ct}_i$ to recover Δ_i
$\bar{\Delta} \leftarrow \mathsf{Median}(\{\Delta_i\})$ ▷ or Multi-Krum; not plaintext mean alone
$w^{(t+1)} \leftarrow w^{(t)} + \bar{\Delta}$

D. BFV Aggregation and Threshold Opening

Additive BFV yields $\mathsf{Dec}(\sum_i \mathsf{Enc}(m_i)) = \sum_i m_i$ under a noise budget growing roughly as $N \cdot O(\sigma\sqrt{n})$. Full-vector packing encrypts all d coordinates as $\lceil d/n \rceil$ independent ciphertexts under the same pk; the ZKP covers the concatenated quantized vector, so partial packing cannot hide an unprotected suffix. After key generation, secret coefficients are Shamir-shared and the monolithic sk is erased. Opening uses Lagrange-weighted partial decryptions with smudging: party i returns $\mu_i = c_1 \star (\lambda_i \cdot s_i)$ (plus noise), and t such shares reconstruct the plaintext without assembling s. Modular negacyclic multiplication with parallel per-chunk encryption keeps full-vector runs interactive on CPU for $d{\approx}5{\times}10^4$. An optional TenSEAL/SEAL sidecar runs in the same process as threshold BFV (`FusedSealThresholdHE`): primary open remains partial-decrypt, while SEAL provides a certified encoder path and consistency check.

a) Post-ZKP aggregation.: Let S be the set of clients that pass Verify. Default aggregation is the coordinate-wise median $\bar{\Delta}_j = \mathrm{med}\{\Delta_{i,j} : i \in S\}$. Multi-Krum and plaintext mean are available as overrides. Median is intentionally applied *after* cryptographic accept: it does not replace the ZKP, and the ZKP does not replace median.

Theorem 1 (Composition soundness and threshold privacy)**.** *Assume SIS binding of* Commit*, RLWE semantic security of BFV, and RO modeling of H=*SHA3-256*. Then: (i) honest $\|\tilde{w}\|_2 \le \tau$ proofs are accepted except for rejection-sampling aborts; (ii) an oversized witness is accepted with probability at most $\approx 2^{-r}$ (binary Unruh) plus SIS advantage; (iii) Enc-consistency plus digesting* **ct** *into challenges prevent attaching a valid proof to a mismatched ciphertext; (iv) the aggregator and any $t{-}1$ decryptors learn nothing beyond what t partial opens reveal. Combinatorial Unruh uniqueness/2^{-r} is machine-checked in Lean 4; EasyCrypt theories instantiate H as SHA3-256 with an O2H-style $q(q{+}1)/2^{256}$ term and algebraic Keccak lane lemmas (`formal/run_formal_ci.py`).*

b) Implementation invariants.: The public repository [12] realizes Algorithm 1 with: (i) identical prove/encrypt dimension d; (ii) ciphertext bytes in Unruh challenges; (iii) crypto-only accept (no trusted Boolean); (iv) full-vector packing; (v) Enc-consistency; (vi) fused SEAL+threshold HE; (vii) default median; (viii) one shared SIS matrix per Unruh prover. Scripts pin seeds and emit JSON under `results/`.

V. EVALUATION

A. Synthetic Multi-Baseline Study

Setup. Synthetic 4-class task (1,000 samples, 784 features, Dirichlet α=0.5), N=5, MLP with d=108,996, 10 rounds, seeds $\{42,\ldots,46\}$. Attacks from round 4: (i) large-norm Gaussian; (ii) sign-flip scaled to $\|\cdot\|_2 = \tau$. Baselines: FedAvg, clip@τ, Multi-Krum (f=1), HE-only, ZKP-only, Hybrid+mean (privacy path without robust agg), and Hybrid+median (default composition). Metrics: final accuracy and cryptographic detection rate (fraction of malicious updates rejected by Verify).

Large-norm poisoning collapses FedAvg/HE-only; clipping, Multi-Krum, and ZKP-style filters retain accuracy (Table IV, Fig. 1–2). Sign-flip within τ is accepted by pure ℓ_2 and Hybrid+mean (0% crypto detection, Table I), while Multi-Krum and Hybrid+median recover accuracy by pairwise or coordinate-wise robust aggregation after cryptographic accept. The Hybrid+mean vs. Hybrid+median contrast is the empirical

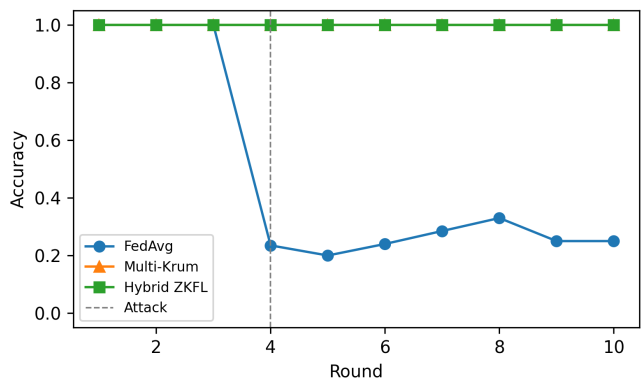

Fig. 1. Synthetic accuracy under large-norm activation (seed 42).

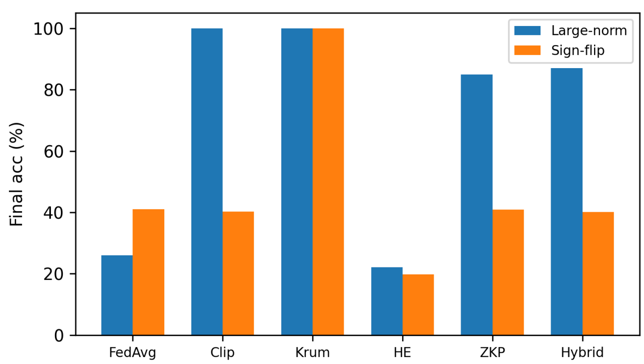

Fig. 2. Final accuracy: large-norm vs. sign-flip (5-seed means).

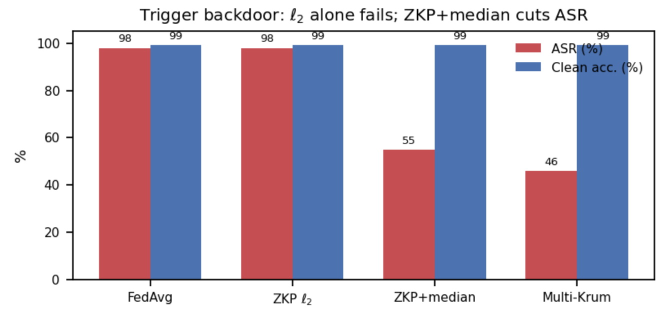

Fig. 3. Backdoor ASR vs. clean accuracy under FedAvg, ℓ_2-ZKP, ZKP+median, and Multi-Krum.

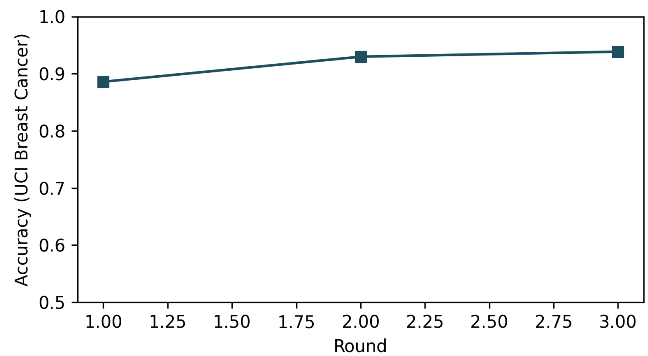

Fig. 4. UCI Breast Cancer accuracy under the full target stack.

punchline for composition: privacy-preserving open without robust aggregation still fails on in-bound flips.

a) Takeaway.: Cryptographic detection is the right metric for oversized attacks; final accuracy after robust aggregation is the right metric for in-bound attacks. Reporting only one of the two understates the threat model in Table I.

B. Scale, Medical, and Backdoor Studies

A compact-model scale study with $N{=}20$ clients and $T{=}30$ rounds reproduces the same qualitative pattern under both attacks: ZKP and hybrid reject oversized updates with high probability, while sign-flip still requires a robust aggregator. Wall-clock remains interactive on CPU for the compact MLP used in that study, which is the intended operating point for validating the crypto pipeline before scaling encoders.

On UCI Breast Cancer Wisconsin (569 samples, 30 clinical features; compact MLP with $d{=}1554$), the target stack—Unruh with $r{=}64$ (library default 128), full-vector fused HE, $(2,3)$ partial threshold decrypt, and post-ZKP median—yields accuracies 88.6% $\rightarrow$ 92.1% $\rightarrow$ 92.1% across three rounds while rejecting a scripted oversized client on the attack round (Fig. 4; JSON in `results/target_protocol_results.json`). Mean round time is $\approx$ 6–10 s on a laptop CPU; fused SEAL consistency error is 0 on the logged rounds. Full-resolution PneumoniaMNIST is evaluated with a compact MLP head ($\approx$12.7k parameters) and with `ConvNet28` ($\approx$51.6k parameters: small CNN stem + FC, no projection, no compact head); a smoke run of the CNN stack reaches $\approx$93.3% accuracy in one round while rejecting an oversized client (`run_medmnist_cnn.py`), with interactive Unruh prove/verify ($\approx$0.1 s at $r{=}4$) and modular BFV encrypt ($\approx$3–4 s) before local training dominates wall-clock.

Trigger backdoors whose poisoned updates remain inside τ are accepted by pure ℓ_2 ZKPs (ASR $\approx$98%, matching FedAvg). Default ZKP+median cuts ASR to $\approx$55% while preserving clean accuracy $\approx$99%; Multi-Krum alone reaches $\approx$46% ASR (Fig. 3). Cryptographic accept filters oversized updates; statistical robust aggregation addresses in-bound poisons. Remaining ASR under median indicates that stronger robust aggregators or DP noise remain complementary—consistent with our non-replacement stance vs. Beskar.

C. Communication Cost

Full-vector HE dominates bandwidth. Uplink scales as $O(N\lceil d/n\rceil n\log q)$ bits plus Unruh $O(r\cdot(d+\ell))$. Table V reports measured artifact payloads; increasing r grows the proof component linearly, while `classic128` ($n{=}4096$)

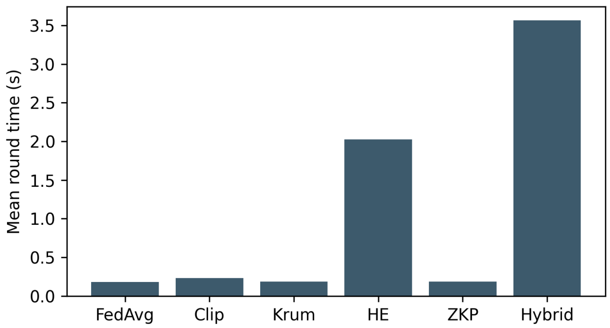

Fig. 5. Mean round time (synthetic hybrid ∼20× FedAvg).

TABLE V
PAYLOAD UNDER UNRUH/n SETTINGS (ARTIFACT ACCOUNTING).

Setting	Msg/round	Notes
Synthetic hybrid	∼100 KB	Reference accounting
UCI target (r=64, n=512)	≈4.7–5.9 MB	Measured JSON
Medical r=128, n=512	higher Unruh share	Linear in r
Medical r=128, n=4096	≈2× ct volume	Classic-128 class
`ConvNet28` smoke (r=4)	≈2.4 MB	d≈51.6k

roughly doubles ciphertext volume for a fixed d. Operators can therefore trade Unruh soundness bits against uplink and CPU without changing the HE packing layout.

VI. SECURITY DISCUSSION

Honest clients with $\|\tilde{w}\|_2 \le \tau$ produce Unruh sessions that pass algebraic and norm checks except for rejection-sampling aborts (bounded retries in the artifact). If $\|\tilde{w}\|_2 \gg \tau$, sessions with challenge bit 1 fail with overwhelming probability under the Gaussian mask design; soundness therefore depends on the fraction of 1-challenges across r repetitions. The library default r=128 targets a ≈ 128-bit combinatorial Unruh class; demos may reduce r for CPU latency at linear proof-size cost. Binding H to ciphertext bytes, together with the Enc-consistency gadget for ρ, prevents a benign proof from being attached to a different ciphertext. BFV semantic security (RLWE) hides individual updates from parties lacking sk; with Shamir (t, n), the aggregator and any $t-1$ decryptors cannot open. We do not claim differential privacy; sites that need DP can compose mechanisms as in Beskar [8].

a) Attack games (informal).: Oversized: adversary wins if Verify $= 1$ on a witness with $\|\tilde{w}\|_2 \gg \tau$. *Split:* adversary wins if a proof for **ct** verifies against $\mathbf{ct}' \neq \mathbf{ct}$. *Curious open:* adversary (aggregator + $<t$ decryptors) wins by recovering an honest client's plaintext update. Theorem 1 bounds the first two under SIS/Unruh/RO and Enc-consistency, and the third under RLWE + Shamir threshold assumptions.

b) Layering.: Table I summarizes the division of labor: ML-KEM for PQ transit, ZKP+binding+Enc-consistency for oversized and split attacks, threshold open for curious aggregators, median for in-bound Byzantine. None of the layers alone meets the joint medical threat model; the contribution is their composition with explicit binding.

c) Deployment.: In a hospital consortium, each site runs a client; a neutral coordinator verifies proofs; t-of-n share holders (for example a DPO and two hospital CISOs) jointly open per-client ciphertexts; median aggregation yields the round update. Daily or weekly training absorbs the observed ∼20× CPU overhead vs. plaintext FedAvg (Fig. 5). For imaging-scale d, HE chunks and Unruh proofs grow with d; shared SIS and modular BFV keep `ConvNet28`-scale prove/encrypt interactive before training dominates. Key ceremony and share custody are operational controls outside the cryptographic core but required for claim (iv).

VII. LIMITATIONS AND REPRODUCIBILITY

Bit-level Keccak-f[1600] carries algebraic lane lemmas (θ-column parity, ρ rotations, π bijection, χ local form, ι locality, packing invertibility, 24-round fold) in EasyCrypt with a FIPS 202 executable checker; a fully discharged algebraic proof of every round identity inside EasyCrypt without the Python oracle remains future work. Full-vector HE+Unruh cost scales with d, but a shared SIS commitment matrix and modular BFV multiplication keep `ConvNet28`-scale runs interactive on CPU. Demo HE uses n=512 (`classic128_demo`); production-oriented n=4096 is available via `ZKFL_HE_PRESET=classic128` and is slower on pure NumPy. Median reduces but does not eliminate backdoor ASR; DP or heavier robust aggregators remain complementary. Seeds, environment overrides (`ZKFL_HE_BACKEND`, `ZKFL_ROBUST_AGG`, `ZKFL_HE_PRESET`), and JSON logs under `results/` reproduce the reported tables; `python formal/run_formal_ci.py` gates the formal surface.

VIII. CONCLUSION

ZKFL-PQ combines ML-KEM-768, ciphertext-bound Unruh proofs with Enc-consistency, threshold BFV partial decryption, and default median aggregation for quantum-resilient medical FL. Experiments on synthetic, tabular, and imaging tasks show that ℓ_2 proofs alone are insufficient against in-bound attacks, while the full stack remains practical on CPU when SIS keys are shared and BFV multiplication is modular. The artifact is intended as a reproducible research stack for hospital-oriented PQ FL, complementary to Beskar-class secure aggregation. Artifact: https://github.com/edlansiaux/pq-zkfl-medical.

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